

Ezra Klukas

UBC ENGINEERING PHYSICS STUDENT · HONOURS MATHEMATICS MINOR

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Education

BASc in Engineering Physics, Honours Mathematics Minor

Vancouver, BC

UBC

Sep. 2023 – Apr. 2028

- Engineering Physics with Honours Mathematics Minor; coursework spanning electronics, robotics, embedded systems, controls, software, and advanced mathematics/physics.
- Self-directed mathematical study alongside coursework in group theory, graph theory, analysis, electrodynamics, quantum mechanics, and systems programming.
- 89.7% cumulative average; awarded TREK Excellence Scholarship for top 5% of domestic students (2024, 2025), Donald J. Evans Scholarship in Engineering, and EXPO 86 Scholarship.

Experience

UBC Sensorimotor Physiology Lab – Dr. Jean-Sébastien Blouin & Jesse Charlton

Vancouver, BC

RESEARCH ENGINEER CO-OP

May 2026 – Aug. 2026

- Building the embedded core of a modular closed-loop human-device experiment platform on an NVIDIA Jetson Orin Nano, integrating high-density EMG acquisition, online motor-unit decomposition, and deterministic EtherCAT servo control.
- Brought up real-time EtherCAT control of a Teknic ClearPath CiA-402 servo in C/C++ with SOEM and IgH, including PDO mapping/remapping, CSP mode, operational-state transitions, distributed-clock synchronization, and cyclic diagnostics.
- Tuned PREEMPT_RT Linux for deterministic control using CPU isolation, IRQ affinity, real-time scheduling, and stress testing; profiled one-hour 1 kHz loop timing, jitter, execution time, and deadline margin under load.
- Developed the offline-to-online EMG software path in MATLAB/C++, including motor-unit filter export, YAML/binary configuration, ring-buffered sample handling, decomposition validation, profiling hooks, and real-time deployment scaffolding.
- Implementing the Intan RHD2000 DDR-SPI reader for timestamped, buffered EMG streaming into peer-designed acquisition.

Siemens Mobility Austria GmbH

Vienna & Graz, Austria

PROJECT MANAGEMENT INTERN

Jul. – Aug. 2023, Jul. – Aug. 2024

- Supported quality engineering and project-management workflows for OBB night-train projects, inspecting reported issues, assessing severity, and coordinating next-step resolution paths.
- Built Python and Office Script automation tools for weekly progress analysis, quality-issue tracking, form generation, and ISO documentation workflows across Austrian and Czech teams.
- Conducted 30+ stakeholder interviews to catalogue internal software tools, superusers, license metadata, and transition requirements, improving continuity during software/process changes.

Selected Projects

Self-driving Autonomous Launching Robot

UBC, ENPH 253

C++/ESP32, KiCad PCB, CAD PROTOTYPING, INSTRUMENTATION, SYSTEMS INTEGRATION

May – Aug. 2025

- Designed and integrated an autonomous competition robot with drivetrain, sensing, pickup, and launching subsystems in a four-person team.
- Designed and soldered H-bridge motor-driver PCBs and debugged power/signal issues using oscilloscopes, DMMs, and systematic subsystem tests.
- Designed a torsion-spring launching arm with a servo-actuated clutch using CAD, 3D printing, machining, and iterative mechanical prototyping.
- Implemented ESP32 C++ firmware for sensing, state-machine control, LiDAR-based behaviours, and system integration.

Simulated Autonomous Driving Competition

UBC, ENPH 353

PYTHON, ROS/GAZEBO, OPENCV, TENSORFLOW/KERAS, LINUX

Sep. – Dec. 2025

- Built a ROS/Gazebo autonomous driving system for a simulated detective robot competition, combining camera-based navigation, traffic-rule logic, and clue-board reading.
- Developed an OpenCV plate-reading pipeline using side cameras, color masks, perspective transforms, contour-based character extraction, and CNN character recognition.
- Implemented fast pixel-mask state-machine driving with NPC detection, plus a lightweight CNN for a visually variable final course section.